

17.4 Empty cells and estimability

Some combinations of levels are never observed, or are impossible. An empty cell's mean does not appear in the distribution of the data, so no unconstrained analysis can estimate it (Proposition 17.1.1(b)). This section works out what survives, building on connectedness (Theorem 8.6.1) and the estimability of interaction contrasts (Proposition 8.6.2).

Throughout, the two-way layout has occupied cells O , with $m = |O|$, every row and every column contains at least one occupied cell, and c is the number of connected components of the design graph of Theorem 8.6.1. A row is **complete** if none of its cells is empty.

17.4.1 What remains estimable

Theorem 17.4.1 — Estimability and tests with empty cells

In the two-way layout described above:

- (a) In the cell-means model, $\sum_i u_i \sum_j w_j \mu_{ij}$ is estimable iff $u_i w_j = 0$ for every empty cell (i, j) . In particular the unweighted mean $\bar{\mu}_i$ is estimable iff row i is complete.
- (b) The estimable interaction contrasts, the tables (d_{ij}) with zero row and column sums and $d_{ij} = 0$ on every empty cell (Proposition 8.6.2(c)), form a space of dimension $m - a - b + c$. The interaction sum of squares $SS(AB \mid \mu, A, B)$ has $m - a - b + c$ degrees of freedom. It addresses the hypothesis that all these contrasts vanish, which holds iff the occupied cell means are additive: $\mu_{ij} = \alpha_i + \beta_j$ for all $(i, j) \in O$ and some α, β .
- (c) If the design is connected, the additive model estimates every cell mean, including those of empty cells, and under additivity $SS(A \mid \mu, B)$ addresses $\alpha_1 = \cdots = \alpha_a$, with $a - 1$ degrees of freedom.
- (d) Code both factors by contrasts and form the interaction columns as products. If at least one row is incomplete and k rows are complete, the last-in sum of squares of A addresses equality of the unweighted means of the complete rows, and it has $\max(k - 1, 0)$ degrees of freedom.

Proof.

- (a) The coefficient of μ_{ij} is $u_i w_j$, and Proposition 17.1.1(b) applies. For $\bar{\mu}_i$ the coefficients are $1/b$ on row i and zero elsewhere.
- (b) Let $T : \mathbb{R}^O \rightarrow \mathbb{R}^a \times \mathbb{R}^b$ send a table $(d_{ij})_{(i,j) \in O}$ to its row sums and column sums. The estimable interaction contrasts form $\mathcal{N}(T)$. The transpose sends $(\mathbf{u}, \mathbf{v})$ to the table $(u_i + v_j)_{(i,j) \in O}$. As in the proof of Theorem 8.6.1, $u_i + v_j = 0$ on O forces $u_i = t_C$ and $v_j = -t_C$ on each component C , with arbitrary t_C . So $\mathcal{N}(T^\top)$ has dimension c , $\text{rank}(T) = a + b - c$, and $\dim \mathcal{N}(T) = m - a - b + c$. The tables $(u_i + v_j)_{(i,j) \in O}$ are exactly the occupied parts of additive tables. They form $\mathcal{C}(T^\top)$, whose orthogonal complement in $\mathbb{R}^O$ is $\mathcal{N}(T)$. So the occupied cell means are additive iff they are orthogonal to every estimable interaction contrast, that is, iff all those contrasts vanish. The interaction sum of squares is $\mathbf{y}^\top (\mathbf{M} - \mathbf{P}_{A+B}) \mathbf{y}$, which addresses the hypothesis that $\mathbf{W}\boldsymbol{\mu}$ lies in the additive model space, that is, that the occupied cell means are additive. Its degrees of freedom are $m - (a + b - c)$.

by Proposition 17.1.1(a) and Theorem 8.6.1(a).

- (c) The first claim is Theorem 8.6.1(c). For the second, let $\mu_{ij} = \alpha_i + \beta_j$. Then $(\mathbf{P}_{A+B} - \mathbf{P}_B)\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$ iff $\mathbf{W}\boldsymbol{\mu} \in \mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$, that is, iff $\alpha_i + \beta_j = \beta'_j$ on O for some β' . Then $\alpha_i + (\beta_j - \beta'_j) = 0$ on O , and in a connected design this forces all α_i to be equal, as in (b). Conversely, equal α_i give such a β' . The degrees of freedom are $\text{rank}[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B] - \text{rank}[\mathbf{1}, \mathbf{Z}_B] = (a + b - 1) - b$.
- (d) As in the proof of Proposition 17.2.3, the full model matrix is $\mathbf{W}$ times the ab independent columns of $\mathbf{K}_A \otimes \mathbf{K}_B$, so its column space is $\mathcal{C}(\mathbf{W})$. The reduced model is $\mathbf{W}\mathbf{F}$. The argument in that proof does not use the counts, and with $\mathbf{w} = \mathbf{1}$ it shows that $\mathcal{C}(\mathbf{F})$ is the space $\mathcal{V}$ of tables with equal unweighted row means. Hence the reduced mean vectors are $\{\mathbf{W}\boldsymbol{\mu} : \boldsymbol{\mu} \in \mathcal{V}\}$: the tables of occupied cell means that can be completed, by choosing values in the empty cells, to a table with equal unweighted row means. Suppose the complete rows share a common unweighted mean t , with t arbitrary if there are none. Each incomplete row can then be given the mean t by choosing the value in one of its empty cells, so the occupied means can be completed. Conversely, a completion cannot change the means of complete rows. So the hypothesis is equality of the unweighted means of the complete rows. It consists of the $k - 1$ restrictions $\bar{\mu}_i = \bar{\mu}_{i_0}$, for the complete rows $i \neq i_0$, which are independent because they involve disjoint sets of cells, and of none if $k \leq 1$. The last-in sum of squares is the drop in residual sum of squares from the reduced model to $\mathcal{C}(\mathbf{W})$, so it addresses this hypothesis (Eq. 6.5.2), and its degrees of freedom are the number of independent restrictions.

■

In part (b), a table with zero row and column sums on the edges of the design graph is a *circulation*. The simplest ones run around cycles with alternating signs. A four-cycle is the familiar tetrad $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl}$; longer cycles give contrasts on six or more cells that survive when every tetrad through them is broken. The number $m - a - b + c$ counts the independent cycles ([Exercise \(C1\)](#)).

Part (d) is the result to remember about Type III. With empty cells it ignores every incomplete row, and it may test nothing at all: an empty cell can hold whatever value makes its row's unweighted mean agree, and such means are not estimable, by (a).

17.4.2 An example with two combinations never grown

Example. (The variety trial with two empty cells)

Suppose that in the trial of Example (Three varieties at three sites) V1 had never been grown at S3, nor V3 at S1. That leaves $n = 33$ plots in $m = 7$ cells, with 26 error degrees of freedom and error mean square 0.3096. The design graph ([Figure 17.4.1](#)) is connected, so the estimable interaction contrasts have dimension $7 - 3 - 3 + 1 = 2$, spanned by two tetrads. The interaction sum of squares is 1.218, $F = 1.97$, $p = 0.160$, and the script checks that the tetrads give the same value through Eq. 17.1.1. The six-cell contrast around V1–S1–V2–S3–V3–S2–V1 is the difference of the tetrads ([Exercise \(B3\)](#)).

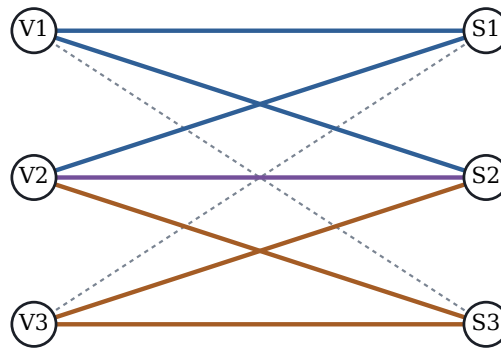


Figure 17.4.1: Design graph of [Example \(The variety trial with two empty cells\)](#) (dashed: empty cells). The blue and orange tetrads share the purple edge V2–S2.

The unweighted means of V1 and V3 are no longer estimable. V1 and V2 share S1 and S2, and the difference of their averages there is 0.603 (standard error 0.268). The additive model is still identified. Its Type II sum of squares for varieties is 5.766 on 2 degrees of freedom, and it fills the empty cells with 8.32 for V1 at S3 and 4.20 for V3 at S1. The first exceeds every observed cell mean: V1's advantage at S1 and S2 is carried to a site where V1 was never grown, although in the generating means it had nearly vanished there (the true mean is 7.6).

The “Type III” sum of squares for varieties is 0 on 0 degrees of freedom: V2 is the only complete row ([Theorem 17.4.1\(d\)](#)). If only V1 at S3 is removed, it is 2.027 on 1 degree of freedom, the sum of squares of $\bar{\mu}_{2.} = \bar{\mu}_{3.}$, for every ordering of the levels.

```
import numpy as np
import pandas as pd
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks (any rank)."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

def rank(*blocks):
    return np.linalg.matrix_rank(np.column_stack(blocks))

missing = ((trial["variety"] == "V1") & (trial["site"] == "S3")) | \
          ((trial["variety"] == "V3") & (trial["site"] == "S1"))
sub = trial[~missing].reset_index(drop=True)          # V1 never grown at S3, V3
```

```

never at S1
y = sub["y"].to_numpy()
n = len(y)
one = np.ones((n, 1))
ZA = pd.get_dummies(sub["variety"]).to_numpy(float)
ZB = pd.get_dummies(sub["site"]).to_numpy(float)
W = np.column_stack([ZA[:, [i]] * ZB for i in range(3)])
m = rank(W) # occupied cells
M, PAB, PA, PB = proj(W), proj(one, ZA, ZB), proj(one, ZA), proj(one, ZB)
sse = y @ (np.eye(n) - M) @ y
df_inter = m - rank(one, ZA, ZB) # m - a - b + 1 for a
connected design
ss_inter = y @ (M - PAB) @ y
F_inter = ss_inter / df_inter / (sse / (n - m))
print(f"occupied cells {m}, interaction df {df_inter}, SS {ss_inter:.3f}, F {F_inter:.2f}")
ss_typeII_A = y @ (PAB - PB) @ y # well defined, a - 1 = 2 df

```

[Run this code online](#)

```

def sum_coding(labels):
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, :-1] - D[:, [-1]]

CA, CB = sum_coding(sub["variety"]), sum_coding(sub["site"])
CAB = np.column_stack([CA[:, [i]] * CB for i in range(2)])
df_III_A = m - rank(one, CB, CAB)
ss_III_A = y @ (M - proj(one, CB, CAB)) @ y
print(f"last-in SS for variety with sum coding: {abs(ss_III_A):.3f} on {df_III_A} df")

```

[Run this code online](#)

17.4.3 How software behaves

A program fitting the rank-deficient effects model picks one coefficient vector and computes its sums of squares from it. The script fits the data of [Example \(The variety trial with two empty cells\)](#) with statsmodels 0.15 under three namings of the levels. It warns that the design is rank-deficient and prints tables anyway. The Type I interaction appears on 4 degrees of freedom instead of 2, with sums of squares 3.96, 2.15 and 2.05 instead of 1.218. The Type II interaction entries are 23.46, 53.21 and 3.17, and the Type III variety entries 9.43, 41.10 and 34.25 on 2 degrees of freedom, where there is nothing to test. The projections give the same answers under every naming, as they must. SAS's "Type IV" sums of squares for empty cells build hypotheses from the occupied cells in a way that depends on the order of the levels (Goodnight 1980). Whatever the program, check that the printed degrees of freedom equal the rank of the projection, and write down the hypothesis.

17.4.4 Recommended practice

1. *Tabulate the counts first*, draw the design graph and count its components. Rows in different components cannot be compared, even under additivity (Theorem 8.6.1(b)).

2. *Ask why the cells are empty.* Cells lost at random fit the theory above. Impossible combinations have no mean, and a main effect averaged over them is fiction. Combinations avoided *because* they were expected to do badly bias every additive fill-in, and no analysis of the observed cells can detect it.
3. *Test the interaction on its $m - a - b + c$ degrees of freedom.* If there are few cycles, additivity is barely checked.
4. *If additivity is credible,* use the additive model and $SS(A \mid \mu, B)$, and flag least squares means that involve empty cells as extrapolations.
5. *If not,* ask estimable questions of the cell means: simple effects, comparisons over shared columns, or a complete sub-table, and state which cells each comparison uses. Do not report a Type III or Type IV table without identifying its hypotheses.

One can also impose just enough interaction constraints to identify the empty cells ([Section 17.3](#)). The resulting estimates are then set by constraints the data cannot test ([Exercise \(B4\)](#)), and should be presented as assumptions.

17.4.5 Exercises

17.4.5.1 A. Check your understanding

Exercise A1.

A 3×4 layout has every cell occupied except $(1, 4)$, $(2, 1)$ and $(3, 1)$. Find m , the number of components of the design graph, the degrees of freedom of the interaction, and the tetrads that are estimable.

Exercise A2.

In [Example \(The variety trial with two empty cells\)](#), which comparisons between V1 and V3 are estimable in the cell-means model? Why is the V1–V2 comparison over S1 and S2 estimable when neither unweighted mean is?

17.4.5.2 B. Practice

Exercise B1.

In a 2×2 layout with cell $(2, 2)$ empty, show that the additive model estimates μ_{22} by $\bar{y}_{21} + \bar{y}_{12} - \bar{y}_{11}$, and find its variance. Show that the last-in sum of squares of A with sum-to-zero coding is identically zero. Compare with [Exercise \(B4\)](#).

Solution. With three occupied cells and a connected graph, the additive model has $1 + 2 + 2 - 1 = 3 = m$ free parameters. Its fitted values are the three cell averages, since $SS(AB \mid \mu, A, B)$ has $3 - 2 - 2 + 1 = 0$ degrees of freedom. Additivity gives $\mu_{22} = \mu_{21} + \mu_{12} - \mu_{11}$, estimated by $\bar{y}_{21} + \bar{y}_{12} - \bar{y}_{11}$ with variance $\sigma^2(1/n_{21} + 1/n_{12} + 1/n_{11})$. Row 1 is the only complete row, so [Theorem 17.4.1\(d\)](#) with $k = 1$ gives zero degrees of freedom, and the sum of squares vanishes for every data set. Directly, the reduced model with columns **1**, the B column and the product column already has rank 3 on the three occupied cells.

Exercise B2.

Take the 4×5 design of Example (Two islands), with occupied cells $(1, 1), (1, 2), (2, 2), (2, 3), (3, 4), (4, 4), (4, 5)$. Find the degrees of freedom of the interaction and the Type II sum of squares of A . Which comparisons of rows can the additive model make?

Exercise B3.

In Example (The variety trial with two empty cells), show that the six-cell contrast $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{23} + \mu_{32} - \mu_{33}$ has zero row and column sums and vanishes on the empty cells. Write it as a combination of the two tetrads, and trace the cycle it corresponds to in Figure 17.4.1.

Solution. Row sums: $1 - 1 = 0$, $-1 + 1 = 0$, $1 - 1 = 0$. Column sums: $1 - 1 = 0$ for S1, $-1 + 1 = 0$ for S2, $1 - 1 = 0$ for S3. It has no coefficient on $(1, 3)$ or $(3, 1)$. With $T_1 = \mu_{11} - \mu_{12} - \mu_{21} + \mu_{22}$ and $T_2 = \mu_{22} - \mu_{23} - \mu_{32} + \mu_{33}$, the contrast is $T_1 - T_2$, because μ_{22} cancels. The cycle is V1–S1–V2–S3–V3–S2–V1. It uses every occupied cell except V2–S2, the edge the two tetrads share.

Exercise B4.

In a 2×3 layout with only cell $(2, 3)$ empty, impose the single constraint $\mu_{12} - \mu_{13} - \mu_{22} + \mu_{23} = 0$. Show that every cell mean becomes estimable and find the estimate of μ_{23} . Can the constraint be tested?

Solution. The constraint gives $\mu_{23} = \mu_{22} + \mu_{13} - \mu_{12}$, so any table satisfying it that vanishes on the occupied cells vanishes everywhere, and Exercise (C2) gives identifiability. The five occupied cell means are unrestricted, so their estimates remain the cell averages, and $\hat{\mu}_{23} = \bar{y}_{22} + \bar{y}_{13} - \bar{y}_{12}$. The constraint involves the empty cell with a nonzero coefficient, so it is not estimable in the cell-means model and cannot be tested (Proposition 11.3.1). The fit is the same whether it holds or not. It only decides what we say about μ_{23} .

17.4.5.3 C. Going deeper**Exercise C1.**

Let F be a spanning forest of the design graph, with $a + b - c$ edges. Show that each occupied cell e not in F closes a unique cycle in $F \cup \{e\}$, and that the alternating-sign contrast $d^{(e)}$ of that cycle is an estimable interaction contrast. Show that the $m - a - b + c$ contrasts $d^{(e)}$ are linearly independent and hence form a basis of the space in Theorem 17.4.1(b).

Solution. Adding an edge $e = (i, j)$ to a forest joins two vertices of the same tree (the graph and the forest have the same components), so it closes exactly one cycle, the tree path from i to j followed by e . A cycle in a bipartite graph has even length and alternates between row and column vertices. Giving its edges alternating signs ± 1 makes the two edges at each vertex cancel, so the row and column sums vanish, and the table is supported on occupied cells. The contrast $d^{(e)}$ is the only one of the family with a

nonzero entry at e , because the other cycles use only forest edges and their own extra edge. So in any vanishing linear combination the coefficient of $d^{(e)}$ is zero, for each e . There are $m - (a + b - c)$ of them, the dimension of the space, so they form a basis.

Exercise C2.

In the trial with only V1 at S3 removed, what does the “Type III” sum of squares for sites test, and on how many degrees of freedom? Verify your answer by modifying the listing.